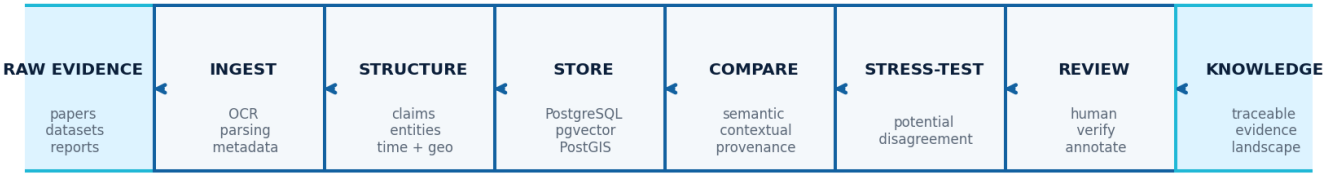
POLARIS-Ω

POLAR Archive with Latent Insight & Scientific Evidence Intelligence

EVIDENCE INTELLIGENCE FOR POLAR RESEARCH

Team: **TechTok**

POLARIS-Ω Evidence Intelligence Pipeline



The workflow is designed to preserve the path from source evidence to reviewed knowledge.

From fragmented archives to traceable scientific knowledge.

PROBLEM	CORE IDEA	FLAGSHIP
Fragmented papers, datasets, reports and institutional memory.	Represent research as claims + evidence + context + provenance.	Evidence Stress-Test Engine: contextual comparison before interpretation.

MVP: repository • claim extraction • hybrid retrieval • evidence comparison • temporal explorer • human review

This report separates implemented capability, planned validation and future scope so technical claims remain defensible.

01 Executive Summary & Problem

POLARIS-Ω is proposed as an evidence-intelligence layer for polar research. It connects papers, datasets, expedition records and metadata at the level of claims, evidence, context and provenance.

The central problem is not simply information retrieval. Researchers must reconstruct relationships between statements across location, time, season, method, instrument, dataset, spatial scale, variable definition and uncertainty.

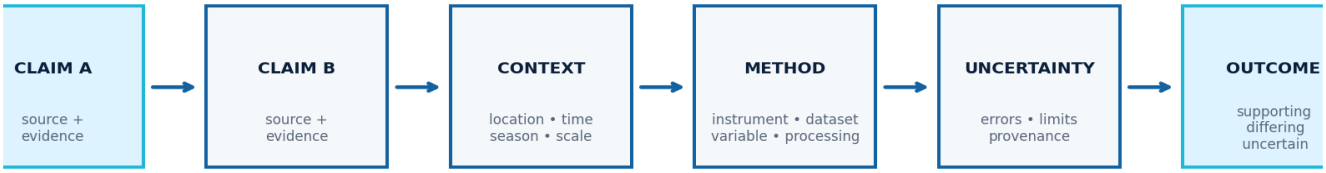
Challenge	What happens today	POLARIS-Ω response
Institutional memory	Field knowledge may remain in reports, notes or experienced researchers.	Structured expedition → method → dataset → publication → lesson relationships.
Sleeping research	Older work can become harder to rediscover as terminology and communities change.	Rediscovery Candidates with transparent signals and human review.
Evidence comparison	Apparently different claims may use different contexts.	Context-aware comparison and stress-test.
Knowledge evolution	New studies add, refine or reinterpret earlier evidence.	Temporal Explorer linked to source evidence.
Provenance	Values can lose their source or processing context.	Source, version, method and transformation traceability.

Design principle: context before conclusion. The system should expose the evidence trail and uncertainty before presenting an interpretation.

The platform is not positioned as a replacement repository, generic chatbot, PDF summarizer or autonomous scientific authority.

02 Proposed Solution & Innovation

Evidence Stress-Test Engine



The engine flags Potential Disagreement; it does not automatically declare a contradiction.

The central workflow is: **RAW EVIDENCE → STRUCTURED CLAIMS → CONTEXT → RELATIONSHIPS → TEMPORAL EVOLUTION → EVIDENCE COMPARISON → EVIDENCE STRESS-TEST → HUMAN REVIEW → TRACEABLE KNOWLEDGE.**

Layer	Mechanism	Output
Source	PDF/data/report ingestion, metadata and identifiers.	Traceable source object.
Claim	Atomic claim extraction with supporting passage.	Structured claim.
Context	Location, time, season, method, instrument, dataset, scale and variable.	Comparable context.
Relationship	Supporting, differing, contextual, uncertain or potential disagreement.	Evidence relationship.
Trust	Provenance + uncertainty + human review state.	Reviewable knowledge.

Innovation is the integration around evidence comparison and traceability. Individual technologies such as embeddings, OCR, NLI and vector search are established; POLARIS-Ω combines them into a polar-research evidence workflow.

The Evidence Gap mechanism is important: when the available context cannot support a reliable comparison, the system identifies what additional evidence or metadata would make the question more answerable.

03 Technical Architecture & Processing

Technical Architecture



Modular services allow processing, storage and inference to scale independently.

Layer	Technology	Purpose
Frontend	Next.js, TypeScript, Tailwind	Research workspace and interaction.
Visuals	D3.js, MapLibre GL JS	Timelines, evidence landscapes and polar maps.
API	FastAPI, Python, REST	Service boundary and business logic.
Workers	Celery, Redis	Asynchronous OCR, extraction and embeddings.
Data	PostgreSQL, pgvector, PostGIS	Metadata, claims, vectors and spatial data.
Storage	S3-compatible object storage	Source files and derived artifacts.
AI/NLP	LLM, embeddings, OCR, NLI	Extraction, retrieval and comparison.
Trust	RBAC, encryption, audit logs	Security and review accountability.

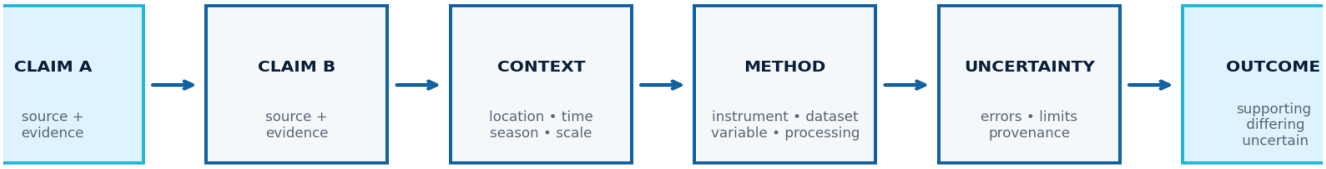
Processing pipeline

Ingestion → parsing/OCR → metadata extraction → claims/entities/time/location/method/instrument extraction → embeddings → hybrid search → contextual comparison → uncertainty/provenance → human review → visualization.

Heavy processing is asynchronous. Source files are separated from operational metadata, and APIs can scale independently. The MVP can use PostgreSQL relationships rather than requiring a separate graph database.

04 Evidence Stress-Test & Human Trust

Evidence Stress-Test Engine



The engine flags Potential Disagreement; it does not automatically declare a contradiction.

Check	Question
Location	Are the claims about comparable places or spatial domains?
Time / season	Do the observation periods and seasons overlap meaningfully?
Method / instrument	Were comparable measurement or estimation procedures used?
Dataset / version	Do both claims use the same or compatible data products?
Scale / variable	Are spatial/temporal scale and operational definitions comparable?
Uncertainty	Do uncertainties overlap, differ or remain unavailable?
Provenance	Can both statements be traced to identifiable source evidence?

Human-in-the-Loop Trust Model



Every insight exposes source, evidence passage, context, uncertainty, provenance and review status.

Trust states: AI-EXTRACTED machine-extracted structure; AI-SUGGESTED candidate relationship; HUMAN-VERIFIED researcher-reviewed; UNCERTAIN ambiguity remains.

05 Provenance, Reproducibility & Data Model

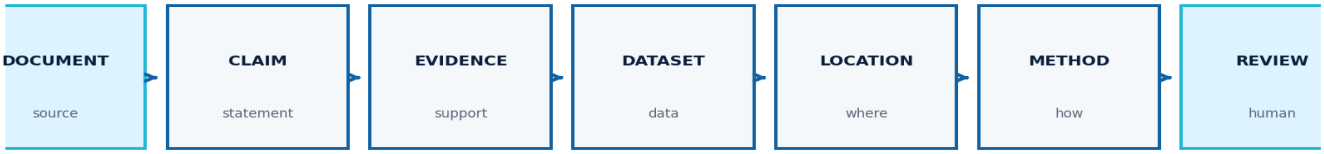
Traceable Provenance Chain



Conceptually informed by W3C PROV: entity, activity, agent, derivation, attribution and versioning.

The provenance design is conceptually informed by W3C PROV. Each important evidence object should retain source, creator, date, instrument, location, method, processing, version, identifier, related publication and review state.

Core Evidence Data Model



Core relationships connect documents, claims, evidence, datasets, questions, locations, methods and review decisions.

Entity	Key relationships
Document	MAKES_CLAIM → Claim; USES_DATASET → Dataset; AUTHORS → Scientist.
Claim	SUPPORTED_BY → Evidence; RELATES_TO → ResearchQuestion; DIFFERS_FROM → Claim.
Evidence	DERIVED_FROM → Source; LOCATED_AT → Location; USES_METHOD → Method.
Review	VERIFIES → Claim/Evidence; records reviewer, status and timestamp.

06 Research Workspace & User Journeys

Workspace	Researcher action	Evidence/trust element
Research Dashboard	Start from question, topic or recent evidence.	Source-linked status.
Polar Map	Explore stations, datasets and evidence by geography.	Coordinate provenance.
Claim Explorer	Search atomic claims and filter by context.	Original passage.
Evidence Comparison	Inspect two or more claims side-by-side.	Context dimensions.
Stress-Test	Investigate Potential Disagreement.	Reason + Evidence Gap.
Temporal Explorer	Trace research across time.	Indexed-source boundaries.
Dataset Explorer	Connect data products and publications.	Version/provenance.
Review Queue	Accept, reject or annotate AI output.	Named review state.
Question Tracker	Follow an open scientific question.	Claims + subquestions.

Learning modes

Beginner	Undergraduate	Researcher
Progressive explanation; source always available.	Concept definitions + structured evidence.	Dense metadata, methods, uncertainty and provenance.

The interface should make the evidence trail visible without forcing users to understand the underlying AI pipeline.

07 Antarctic Sea-Ice Demonstration Case Study

Antarctic sea-ice research is useful for demonstrating contextual evidence comparison because analyses may differ by region, season, time window, dataset, variable and physical mechanism.

Step	System action	Judge-visible result
1	Retrieve relevant claims.	Claim cards with source.
2	Normalize context.	Location/time/method panel.
3	Identify apparently different statements.	Potential Disagreement.
4	Stress-test context.	Dimension-by-dimension comparison.
5	Inspect evidence.	Passage/data/source trace.
6	Classify relationship.	Supporting / differing / contextual / uncertain.
7	Find missing context.	Evidence Gap.
8	Researcher review.	Human-verified or rejected relationship.

Research basis

The demonstration can use NSIDC Sea Ice Index documentation and peer-reviewed studies including Yu et al. (2023), Jena et al. (2024), Wang et al. (2024) and Espinosa et al. (2024). These works examine Antarctic sea-ice variability, regional/seasonal behaviour and the unusual 2023 low-ice conditions from different perspectives.

The report deliberately does not label those papers as contradictory. Their differing contexts demonstrate why scientific comparison must inspect time, geography, methods, datasets and uncertainty before drawing conclusions.

Expected demo moment: a user sees two apparently different claims → opens the comparison → the system explains the contextual dimensions → source passages are shown → an Evidence Gap or human-review action appears.

08 Existing Ecosystem & Positioning

Existing platform / source	Primary role	Relationship to POLARIS-Ω
NCPOR / NPDC	Indian polar research and data infrastructure.	Authoritative source ecosystem.
NSIDC	Sea-ice data products and documentation.	Long-term dataset source.
NASA Earthdata	Earth-observation discovery and access.	Remote-sensing source.
BAS / UK Polar Data Centre	Polar data and institutional collections.	Institutional data source.
NSF Arctic Data Center	Arctic data/software preservation and provenance.	Arctic source.
Polar Data Catalogue	Polar metadata and data discovery.	Cross-source reference.
Antarctic Treaty Secretariat	Antarctic information exchange and records.	Institutional document source.

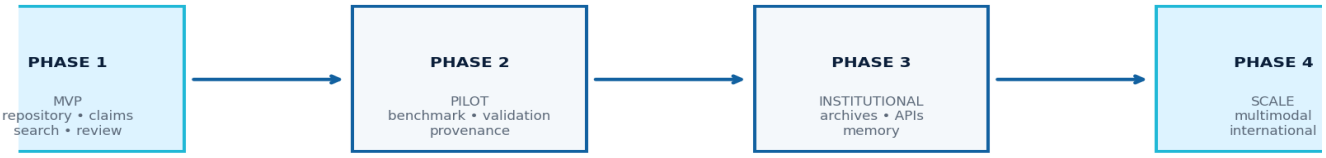
Positioning

Existing system	POLARIS-Ω adds
Repository / catalogue	Claim-level evidence representation and relationships.
Search engine	Context-aware comparison and temporal/spatial exploration.
RAG assistant	Persistent evidence objects, provenance and review state.
Knowledge graph	Source passages, uncertainty and human verification around relationships.
Dashboard	Research-question and evidence workflows rather than only metrics.

POLARIS-Ω complements authoritative data services; it is not presented as a replacement for them.

09 Feasibility, Viability & Roadmap

Implementation Roadmap



Each phase produces a testable increment; advanced capabilities remain separate from the MVP.

Phase	Scope	Gate
1 — MVP	Repository, ingestion, claims, hybrid search, comparison, stress-test, temporal explorer, review.	Traceability works end-to-end.
2 — Pilot	Annotated benchmark, provenance, rediscovery candidates, feedback.	Measured extraction/comparison performance.
3 — Institutional	Archive/data connectors, RBAC, APIs, institutional memory.	Governance and integration checks.
4 — Scale	Multimodal evidence and broader integrations.	Interoperability and domain validation.

Scalability

Use asynchronous workers for expensive jobs, object storage for large artifacts, PostgreSQL indexing for structured queries, pgvector for semantic retrieval, PostGIS for spatial operations and Redis for queues/caching.

Viability

The architecture uses mature components and allows incremental delivery. Advanced multimodal, international federation and autonomous reasoning are intentionally separated from the MVP.

10 Validation, Security & Responsible AI

Evaluation	Method	Metric
Claim extraction	Annotated claim spans.	Precision / Recall / F1.
Entity & temporal extraction	Annotated people, datasets, places, dates and methods.	Precision / Recall / F1.
Evidence linkage	Verify source passage/object for each claim.	Link accuracy / coverage.
Comparison	Expert-labelled relationship set.	Precision / Recall / F1.
False disagreement	Check Potential Disagreement flags.	False-positive rate.
Human review	Researcher accepts/corrects/rejects.	Acceptance / correction / rejection.
Traceability	Audit displayed insights.	Traceability coverage.

Benchmark plan

Start with approximately 50–100 manually annotated documents spanning papers, reports and data documentation. Use a held-out evaluation set and record model, prompt and preprocessing versions. Replace every planned metric with measured results before claiming performance.

Risk controls

Risk	Mitigation
Hallucination	Retrieval grounding, source passage, uncertainty and review.
False disagreement	Context stress-test + human verification.
OCR / metadata errors	Confidence, original page access and manual correction.
Dataset version conflict	Version-aware provenance.
Licensing / access	Respect source licences and permissions.
Security	RBAC, encryption, audit logs and controlled access.

11 Impact, References & Judge Readiness

Stakeholder	Value
Polar researchers	Faster evidence reconstruction and contextual comparison.
Data curators	Links between datasets, publications, metadata and provenance.
Expedition teams	Structured institutional memory when verified records exist.
Students	Evidence-backed learning with progressive explanation.
Institutions	Reusable evidence layer across scientific programmes.
Interdisciplinary teams	Cross-domain discovery through shared entities, time and location.

Research basis and official sources

NCPOR / National Polar Data Center; NSIDC Sea Ice Index; NASA Earthdata; British Antarctic Survey / UK Polar Data Centre; NSF Arctic Data Center; Polar Data Catalogue; Antarctic Treaty Secretariat; W3C PROV and PROV-O.

Key papers: Yu et al. (2023), International Journal of Climatology, DOI 10.1002/joc.8010; Jena et al. (2024), Geophysical Research Letters, DOI 10.1029/2023GL107561; Wang et al. (2024), Communications Earth & Environment; Espinosa, Blanchard-Wrigglesworth & Bitz (2024), Communications Earth & Environment.

Judge Q&A;

Question	Answer
Why not RAG?	Persistent claims, evidence, context, provenance, relationships and review state.
How avoid hallucination?	Source grounding + evidence passage + uncertainty + human verification.
How handle contradictions?	Potential Disagreement → contextual stress-test → human interpretation.
What is MVP?	Repository + claims + search + comparison + stress-test + temporal + review.
How validate?	Annotated benchmark with extraction, comparison and human-review metrics.
Can it scale?	Asynchronous processing + object storage + PostgreSQL/pgvector/PostGIS + scalable services.

FINAL PROPOSITION

POLARIS-Ω turns decades of fragmented polar research into a traceable evidence landscape — helping researchers discover, compare and understand how scientific knowledge evolves, while keeping final judgment with humans.

Submission checklist: insert actual prototype screenshots; replace planned metrics with measured results; verify every DOI/URL; add final YouTube/report links; attribute external material; clearly label MVP vs future scope.